

**Department of Electronics and Communication Engineering**
Academic Year 2021 – 2022 (Odd Semester)**Degree, Semester & Branch: B.E., III & ECE B****Course Code & Title: EC8352 & Signals and Systems****Name of the Faculty member (s): Ms.S.Harini Shriram****Innovative Practice Description**

- **Unit / Topic:** II/ Laplace Transform-Properties
- **Course Outcome:** CO : The students will be able to Apply Fourier Series, Laplace transform and Fourier transform in signal analysis
- **Topic Learning Outcome:** TLO: Analyze CT aperiodic signals using Laplace transform and apply its properties
- **Activity Chosen:** Flipped class room
- **Justification:** It allows students to learn in their own pace, it encourages students to actively engage with lecture material, it frees up actual class time for more effective, creative and active learning activities and students take control and responsibility for their learning.
- **Time Allotted for the Activity:** 40 minutes
- **Details of the Implementation:**
Specific topic was given to the students learn on their own. Resources like reference book, videos were given to the students. Students are asked to prepare more in depth than before. Students are separate into groups where students are given a task to perform. Get the class back together to share the individual group's work with everyone.

• **CO – PO / PSO mapping:**

CO	PO1	PO2	PO5	PO8	PO9	PO10	PSO1
CO2	3	2	2	2	3	2	3

• **PO / PSO mapped:**

Innovative practice	PO8	PO9	PSO1
	2	1	3
Justification for correlation	The course outcome is moderately correlated, since the Flipped Classroom activity involves presentations which apply moral & ethical principles while collecting the materials related to the topic.	A collaborative activity (flipped classroom) is planned where the students work as a member of a team to achieve the goal. So, the course outcome is highly correlated.	The students will be able to innovate new ideas to solve existing/novel problems in the analysis of CT signals. So, the PSO1 is mapped at level 3

- **Images / Screenshot of the practice:**



- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

Self-learning and accountability of the students got improved. They learn how to communicate with team members and work together to achieve common goals.

- ❖ ***Benefit of the practice:***

Every student got equal opportunity to take part in this activity. The success of the activity was evaluated by asking the same question in Internal Assessment test II – Around 80% of students answered correct

- ❖ ***Challenges faced in implementation:***

The main challenge faced is that few students not exposed to flipped class room. Students struggle with self-discipline and may turn up to class without having absorbed the lesson. Few students hesitated to come forward and to interact. I have encouraged them to come forward and share their points

References:

- ❖ https://www.rcboe.org/cms/lib/GA01903614/Centricity/Domain/15451/Flip_your_Classroom.pdf
- ❖ Bergmann, J., & Sams, A. (2012). Flip your classroom: Reach every student in every class every day. Eugene, Or: International Society for Technology in Education.
- ❖ Center for Teaching Innovation at Cornell University. (2017). Flipping the classroom. Retrieved from <https://www.cte.cornell.edu/teaching-ideas/designing-your-course/flipping-the-classroom.html>.



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Innovative Practice Description

- **Unit / Topic:** III/ Differential Equation- Laplace transforms in Analysis of CT systems- Total Response
- **Course Outcome:** CO: The students will be able to compute the output of Linear Time Invariant Continuous Time Systems using Laplace and Fourier Transforms.
- **Topic Learning Outcome:** TLO: Analyze differential equation relating the input & output of a LTI-CT system using Laplace transform and obtain its response.
- **Activity Chosen:** Round table activity
- **Justification:** Roundtable is a good cooperative structure and interactive activity. This method bring about both individual accountability and achievement of the team goals. In this activity students depend on each other to succeed.
- **Time Allotted for the Activity:** 40 minutes

- **Details of the Implementation:**

Each team consists of five or six students. Each group is assigned a different set of problems to work on. Everyone in the group discussed the topic and came up with a solution. Then there would be a group discussion so that everyone is aware of all those problems.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO5	PO8	PO9	PO10	PSO1
CO3	3	2	2	2	3	2	3

- **PO / PSO mapped:**

Innovative practice	PO8	PO9	PSO1
	2	3	3
Justification for correlation	The PO8 is mapped at level 2 since the Collaborative learning round table activity is planned here so the students must follow basic ethical practice among themselves and within groups.	It is highly correlated since the students work well as individuals and as a team while doing the round table activity.	Students will be able to come up with new ways to solve existing/new problems while determining the response to an LTI CT system. As a result, the PSO1 is mapped at level 3

- **Images / Screenshot of the practice:**



- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

Students felt it was very interesting and useful. They learn how to communicate with team members and work together to achieve common goals.

- ❖ ***Benefit of the practice:***

Every student got equal opportunity to take part in this activity and they learned all the variety of problem with in the short span of time. The success of the activity was evaluated by asking the same question in Internal Assessment test II – Around 80% of students answered correct

- ❖ ***Challenges faced in implementation:***

The main challenge faced is that few students hesitated to ask their doubt openly in the group.

References:

- ❖ <https://www.modelteaching.com/education-articles/teaching-strategies/re-energizing-classroom-discussion-through-round-table-circles>
- ❖ <https://files.eric.ed.gov/fulltext/EJ1152588.pdf>

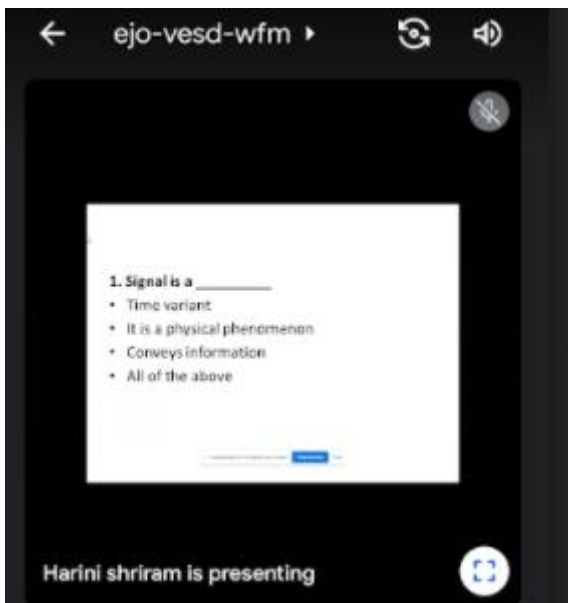
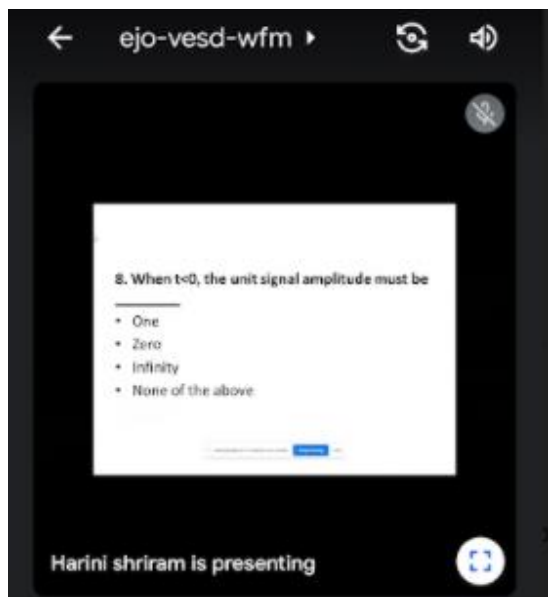
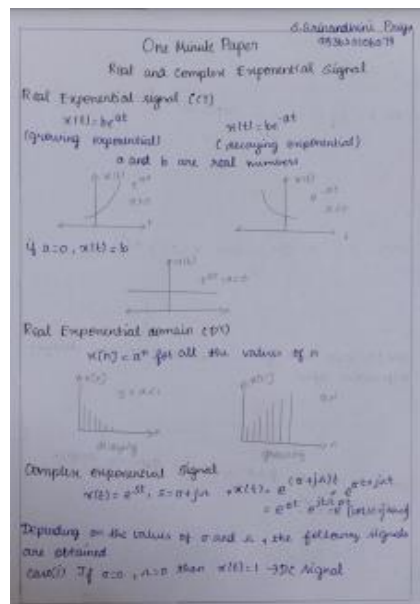
RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2021 - 2022 (ODD Semester)

INNOVATIVE TEACHING METHOD PLAN

Degree, Semester & Branch: III Semester B.E.ECE B

Course Code & Title: EC8352 & Signals and Systems

Name of the Faculty member: Ms.S.Harini Shriram

UNIT I			
1.	18.08.21	Introduction to Signals and systems standard signals –step, ramp	Quesscussion
			
2	19.08.21	Standard signals-Real and complex exponentials and Sinusoids	Minute Paper
			

3	07.09.21	Classification of systems -Causal & Non-causal, Stable & Unstable	Crowd Sourcing
			

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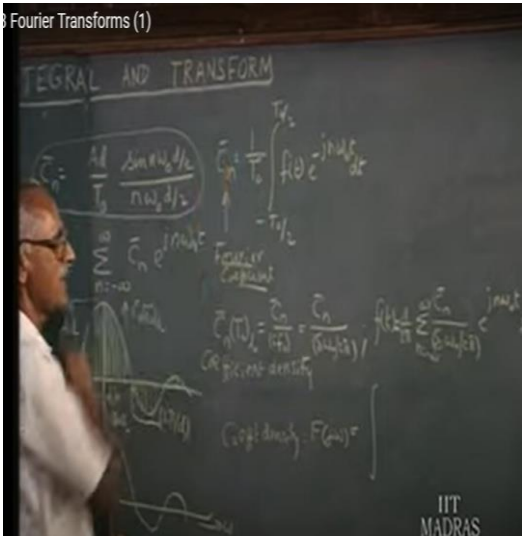
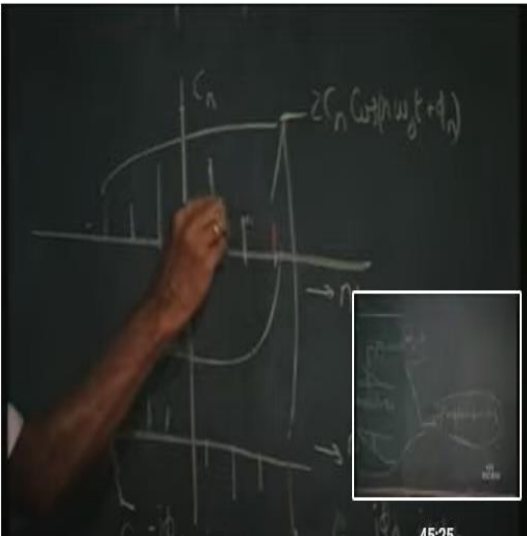
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UNIT II

1.	15.09.21	Fourier series for periodic signals – Exponential Fourier series	Active learning-Clickers
			
2	16.09.21	Fourier series for periodic signals – Properties of Fourier Series -	Open book test(library)
			

3	21.09.21	Fourier Transform - Properties	NPTEL video
			

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UNIT III

1.	30.09.21	Convolution integrals	Online Simulation
			
2	04.10.21	Convolution integrals - Properties	Head and Talk
			

3	12.10.21	Differential Equation- Laplace transforms in Analysis of CT systems-Total Response	Round Table
			

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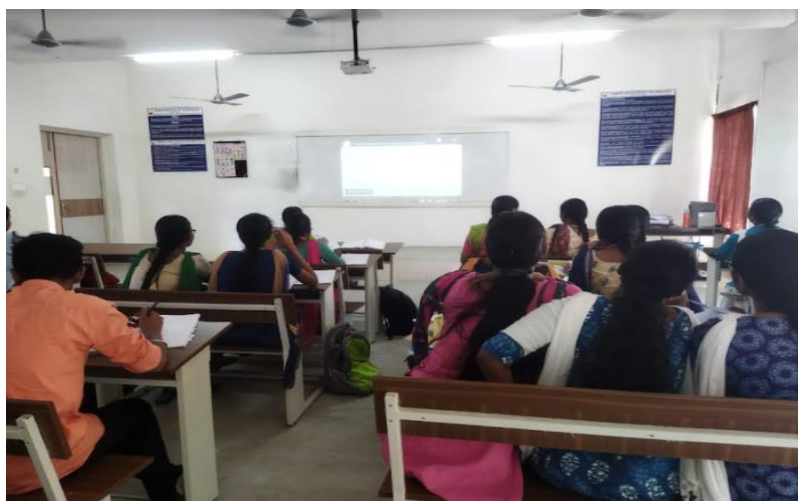
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UNIT IV

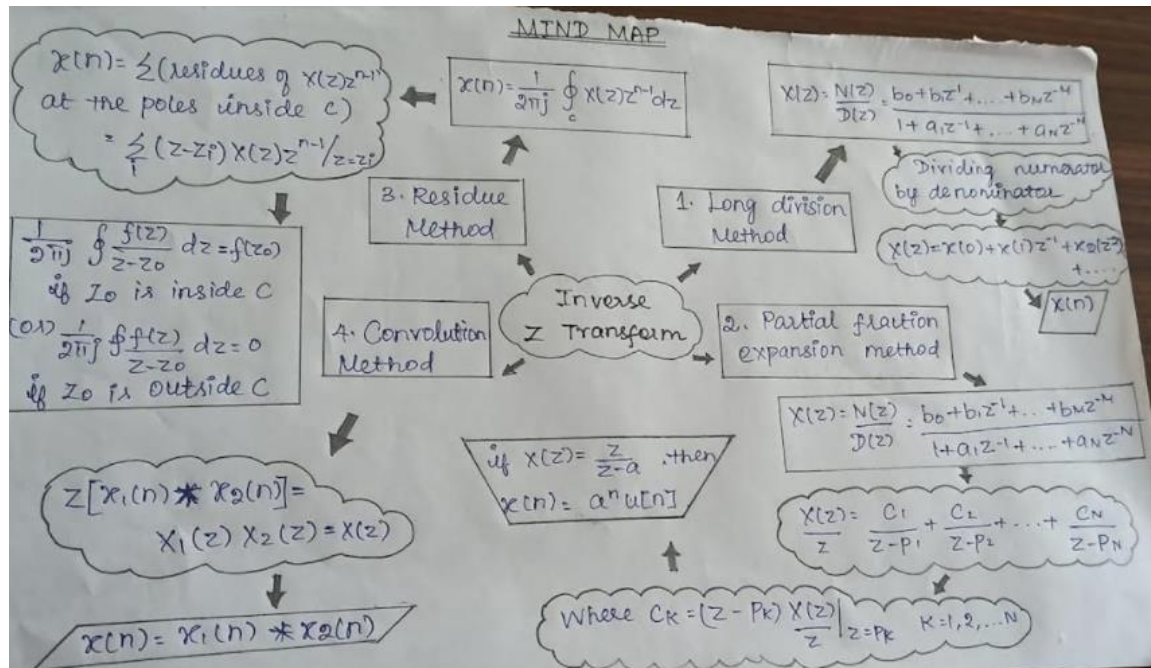
1.	21.10.21	Introduction, Baseband signal Sampling	Online Simulation
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Sampling and Reconstruction: <https://www.youtube.com/watch?v=b-zbYaTPmkE>
https://www.youtube.com/watch?v=CxP_-zAo89E



2	28.10.21	Properties of DTFT	Open book test
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UNIT V

1.	18.11.21	Convolution sum	Online Simulation
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<https://www.youtube.com/watch?v=nckLTwWeVKw>



2	19.11.21	Convolution sum	Sorting strips
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3	26.11.21	Z Transform in analysis of Recursive & Non-Recursive systems	Finger Signal
